



Related Rates with Implicit Differentiation

Calculus Worksheet · Grade 11-12

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Learning Objectives

- Apply implicit differentiation with respect to time to geometric formulas
- Use the product rule when differentiating related rates expressions
- Solve for an unknown rate by substituting given values into a derived equation

For each problem, differentiate the given formula implicitly with respect to time, then substitute the given values to find the requested rate.

1. Given the area of a circle, find dA/dt when $r=2$ and $dr/dt=3$.

$$A = \pi r^2, r = 2, \frac{dr}{dt} = 3$$

Answer: _____

2. Given the lateral surface area of a cylinder, find dr/dt when $r=2$, $h=4$, $dA/dt=16\pi$, and $dh/dt=2$.

$$A = 2\pi rh, r = 2, h = 4, \frac{dA}{dt} = 16\pi, \frac{dh}{dt} = 2$$

Answer: _____

3. The radius of a circle is increasing at 5 cm/s. Find the rate of change of the area when $r=10$ cm.

$$A = \pi r^2, r = 10, \frac{dr}{dt} = 5$$

Answer: _____

4. The volume of a sphere $V = (4/3)\pi r^3$ is changing. Find dV/dt when $r=3$ and $dr/dt=2$.

$$V = \frac{4}{3}\pi r^3, r = 3, \frac{dr}{dt} = 2$$

Answer: _____

5. For a square with side s , find dA/dt when $s=6$ and $ds/dt=4$.

$$A = s^2, s = 6, \frac{ds}{dt} = 4$$

Answer: _____

6. A cylinder has volume $V = \pi r^2 h$. Find dV/dt when $r=2$, $h=5$, $dr/dt=1$, and $dh/dt=3$.

$$V = \pi r^2 h, r = 2, h = 5, \frac{dr}{dt} = 1, \frac{dh}{dt} = 3$$

Answer: _____



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7. A right triangle has legs x and y with $x^2 + y^2 = z^2$. Find dz/dt when $x=3$, $y=4$, $dx/dt=2$, and $dy/dt=1$.

$$x^2 + y^2 = z^2, \quad x = 3, \quad y = 4, \quad \frac{dx}{dt} = 2, \quad \frac{dy}{dt} = 1$$

Answer: _____

8. The area of a rectangle is $A=lw$. Find dA/dt when $l=8$, $w=5$, $dl/dt=3$, and $dw/dt=2$.

$$A = lw, \quad l = 8, \quad w = 5, \quad \frac{dl}{dt} = 3, \quad \frac{dw}{dt} = 2$$

Answer: _____

9. The surface area of a sphere is $S=4\pi r^2$. Find dS/dt when $r=4$ and $dr/dt=2$.

$$S = 4\pi r^2, \quad r = 4, \quad \frac{dr}{dt} = 2$$

Answer: _____





Remind students to differentiate first symbolically before substituting numerical values, and to apply the product rule when two variables both change with time.

Solutions

1. Given the area of a circle, find dA/dt when $r=2$ and $dr/dt=3$.

$$A = \pi r^2, \quad r = 2, \quad \frac{dr}{dt} = 3$$

- Differentiate both sides with respect to time, treating π as a constant.
- The derivative of r squared with respect to time is two r times dr/dt .
- Substitute r equals two and dr/dt equals three into the equation.
- Multiply two times two times three times π to get twelve π .

Answer: $\frac{dA}{dt} = 12\pi$

2. Given the lateral surface area of a cylinder, find dr/dt when $r=2$, $h=4$, $dA/dt=16\pi$, and $dh/dt=2$.

$$A = 2\pi rh, \quad r = 2, \quad h = 4, \quad \frac{dA}{dt} = 16\pi, \quad \frac{dh}{dt} = 2$$

- Differentiate A equals two π r h with respect to time using the product rule.
- Obtain dA/dt equals two π h dr/dt plus two π r dh/dt .
- Substitute dA/dt equals sixteen π , h equals four, r equals two, and dh/dt equals two.
- Simplify to sixteen π equals eight π dr/dt plus eight π .
- Subtract eight π from both sides and divide by eight π to get dr/dt equals one.

Answer: $\frac{dr}{dt} = 1$

3. The radius of a circle is increasing at 5 cm/s. Find the rate of change of the area when $r=10$ cm.

$$A = \pi r^2, \quad r = 10, \quad \frac{dr}{dt} = 5$$

- Differentiate area with respect to time to get dA/dt equals two π r dr/dt .
- Substitute r equals ten and dr/dt equals five.
- Multiply two times ten times five times π .
- The result is one hundred π square centimeters per second.

Answer: $\frac{dA}{dt} = 100\pi \text{ cm}^2/\text{s}$

4. The volume of a sphere $V = (4/3)\pi r^3$ is changing. Find dV/dt when $r=3$ and $dr/dt=2$.

$$V = \frac{4}{3}\pi r^3, \quad r = 3, \quad \frac{dr}{dt} = 2$$

- Differentiate V with respect to time to obtain dV/dt equals four π r squared dr/dt .
- Substitute r equals three and dr/dt equals two.
- Compute four times nine times two to get seventy-two.
- Therefore dV/dt equals seventy-two π .

Answer: $\frac{dV}{dt} = 72\pi$



5. For a square with side s , find dA/dt when $s=6$ and $ds/dt=4$.

$$A = s^2, s = 6, \frac{ds}{dt} = 4$$

→ Differentiate area equals s squared with respect to time.

→ Obtain dA/dt equals two s ds/dt .

→ Substitute s equals six and ds/dt equals four.

→ Multiply two times six times four to get forty-eight.

Answer: $\frac{dA}{dt} = 48$

6. A cylinder has volume $V = \pi r^2 h$. Find dV/dt when $r=2$, $h=5$, $dr/dt=1$, and $dh/dt=3$.

$$V = \pi r^2 h, r = 2, h = 5, \frac{dr}{dt} = 1, \frac{dh}{dt} = 3$$

→ Differentiate V with respect to time using the product rule.

→ Get dV/dt equals two $\pi r h$ dr/dt plus πr squared dh/dt .

→ Substitute r equals two, h equals five, dr/dt equals one, and dh/dt equals three.

→ Compute twenty π plus twelve π to obtain thirty-two π .

Answer: $\frac{dV}{dt} = 32\pi$

7. A right triangle has legs x and y with $x^2 + y^2 = z^2$. Find dz/dt when $x=3$, $y=4$, $dx/dt=2$, and $dy/dt=1$.

$$x^2 + y^2 = z^2, x = 3, y = 4, \frac{dx}{dt} = 2, \frac{dy}{dt} = 1$$

→ Differentiate both sides with respect to time.

→ Obtain two x dx/dt plus two y dy/dt equals two z dz/dt .

→ Find z from three squared plus four squared equals twenty-five, so z equals five.

→ Substitute values to get six times two plus eight times one equals ten dz/dt .

→ Solve to get dz/dt equals two.

Answer: $\frac{dz}{dt} = 2$

8. The area of a rectangle is $A=lw$. Find dA/dt when $l=8$, $w=5$, $dl/dt=3$, and $dw/dt=2$.

$$A = lw, l = 8, w = 5, \frac{dl}{dt} = 3, \frac{dw}{dt} = 2$$

→ Differentiate A equals lw with respect to time using the product rule.

→ Obtain dA/dt equals dl/dt times w plus l times dw/dt .

→ Substitute l equals eight, w equals five, dl/dt equals three, and dw/dt equals two.

→ Compute three times five plus eight times two equals fifteen plus sixteen, giving thirty-one.

Answer: $\frac{dA}{dt} = 31$

9. The surface area of a sphere is $S=4\pi r^2$. Find dS/dt when $r=4$ and $dr/dt=2$.

$$S = 4\pi r^2, r = 4, \frac{dr}{dt} = 2$$

→ Differentiate S equals four π r squared with respect to time.

→ Obtain dS/dt equals eight πr dr/dt .

→ Substitute r equals four and dr/dt equals two.

→ Compute eight times four times two times π to get sixty-four π .

Answer: $\frac{dS}{dt} = 64\pi$

